Rejection Sampling for Medical MCQ Generation and Fine-Tuning TinyLlaMA for Layperson QA

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Hi, I'm Chelsea Ramos.

Today I'll share my project on improving layperson medical question answering using rejection sampling and fine-tuning TinyLlaMA.

Reliable medical answers should be easy to access, but general models often miss the mark.

This project hopes to tackles that gap.

Introduction

First, I'll walk through some quick background, and what my project set out to do.

Background

- Medical QA: complex, needs accuracy
- Layperson QA: needs simplicity
- LLMs: need domain fine-tuning
- Good data is critical

Medical question answering demands precision and clarity.

For everyday users, answers need to be simple and trustworthy.

Big models like TinyLlaMA aren't tuned for that by default.

And without good, targeted data, even a fine-tuned model won't perform well.

Objectives

- Generate MCQs from MedQuAD
- Filter with rejection sampling
- Fine-tune TinyLLaMA
- Compare to baseline and MedAlpaca

My objectives were to generate multiple-choice questions from MedQuAD,

filter for quality using rejection sampling,

fine-tune TinyLlaMA with supervised learning,

and compare it to both the baseline and MedAlpaca models.

Method

Now I'll dive into the methods —
starting with the dataset and workflow.

MedQuAD Dataset

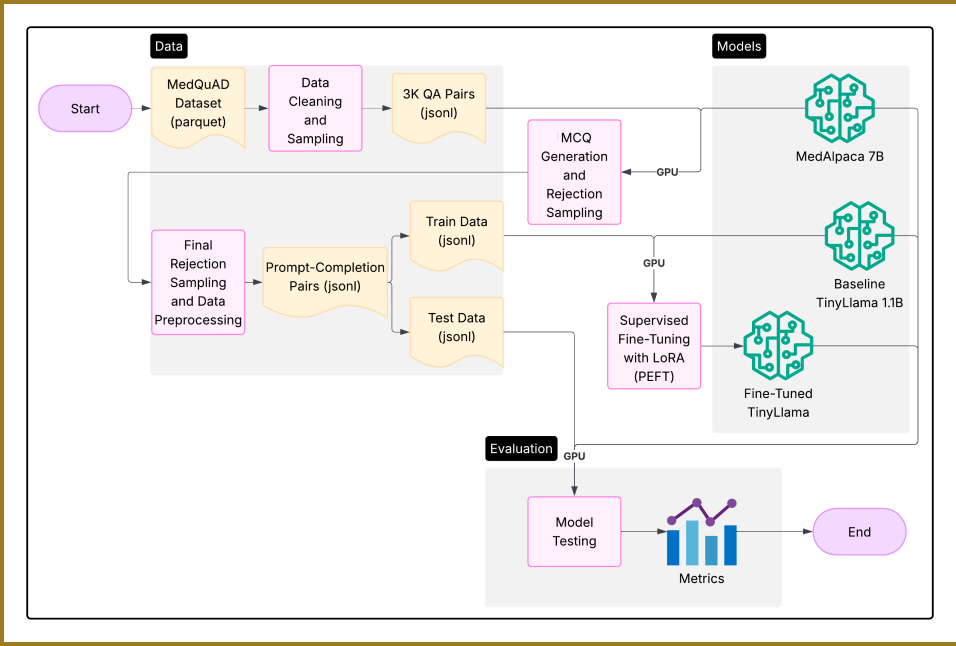
- 47K+ layperson medical QA pairs
- Trusted U.S. health sources
- Clear, accessible answers

I used the MedQuAD dataset — over 47,000 question-answer pairs built for layperson understanding.

Sources include trusted organizations like National Institutes of Health and MedlinePlus.

It was a good fit for creating accessible multiple-choice questions.

Workflow



The pipeline was simple:

sample question-answer pairs, generate multiple-choice questions w/ MedAlpaca, filter using rejection sampling, fine-tune TinyLlaMA, and then evaluate.

MCQ Generation

```
# Few-shot example to encourage structure
few_shot_example = """
Question: What is asthma?
Answer: Asthma is a lung condition that causes airway inflammation and breathing difficulty.
<choices>
A. A skin condition
B. A viral infection
C. A lung disease that causes airway inflammation
D. A type of heart failure
</choices>
<reason>Asthma is a lung condition that causes airway inflammation and breathing difficulty.</reason>
<answer>C</answer>
"""

# Format QA into prompt
def format_prompt(question, answer):
    return f"""
Create layperson multiple-choice answer choices and reasoning for the given medical Q&A pair.
Include 4 total choices: one correct and three plausible, but incorrect. Follow the XML-style structure below:
{few_shot_example}
Now it's your turn:
Question: {question}
Answer: {answer}
"""
```

For generation, I sampled about 3,000 question-answer pairs.

Then, I used a custom few-shot prompt with reasoning to guide the MedAlpaca's multiple-choice question generation.

Generated MCQs


```
{"prompt":"You are a helpful and concise medical assistant.\n\nAnswer the
following multiple-choice question. First, briefly explain your reasoning using
relevant medical knowledge. Then state your final answer letter using exactly one
<answer></answer> tag.\n\nInstructions:\n- Use only one <answer> tag (e.g.,
<answer>B</answer>)\n- Choose the best answer letter\n- Stop immediately after
the </answer> tag\n\n<question>\nWhat causes Gum (Periodontal) Disease?\nA.
Genetic mutations\nB. Poor dental hygiene\nC. Injury to the gums\nD. Periodontal
abscesses\n</question>\n","completion":"Gum disease is caused by dental plaque --
a sticky film of bacteria that builds up on teeth. Regular brushing and flossing
help get rid of plaque. But plaque that is not removed can harden and form tartar
that brushing doesn't clean. Only a dentist or dental hygienist can remove
tartar.\n<answer>B</answer>"}
```


- 511 MCQs finalized
- 408 train / 103 test (80/20 split)

From 3,000 question-answer pairs, I generated 728 multiple-choice questions.

After review, 511 were finalized — about 400 for training and 100 for testing.

Results

Now, let’s jump into the results.

Loss

Small dataset: quick, noisy training

[48/48 06:11, Epoch 3/3]

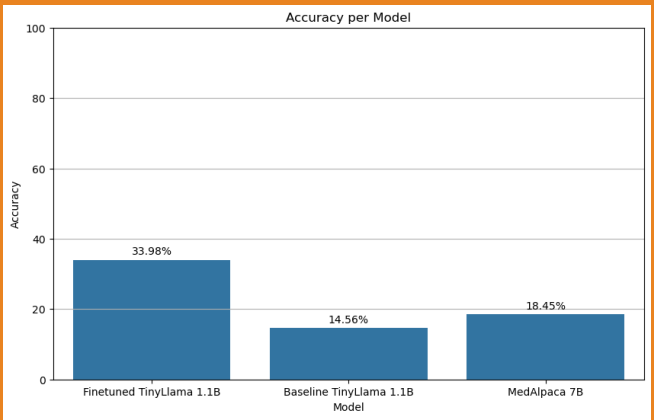
Epoch	Training Loss	Validation Loss
1	0.846100	0.489880
2	0.370500	0.475589
3	0.363300	0.481885

Since the dataset was small, training finished fast.

The loss curves were a little noisy, but overall showed improvement.

Accuracy

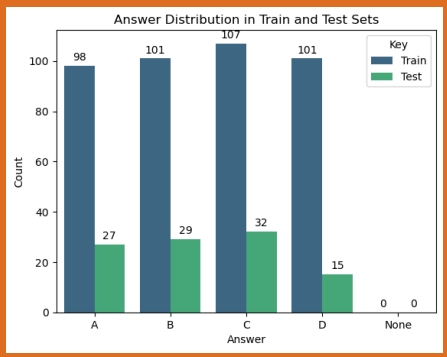
Finetuning improved accuracy, but gains were modest

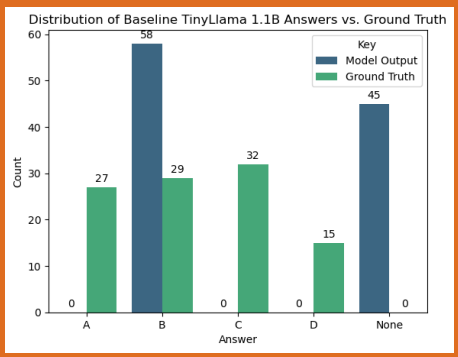


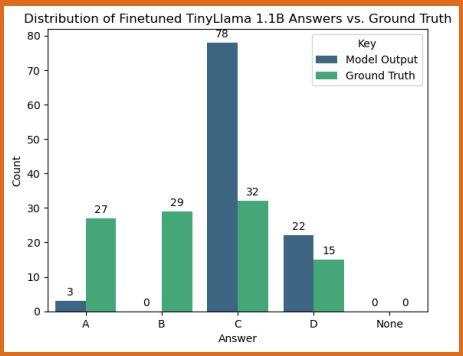
Fine-tuning gave a clear accuracy boost over both the baseline TinyLlaMA and MedAlpaca.

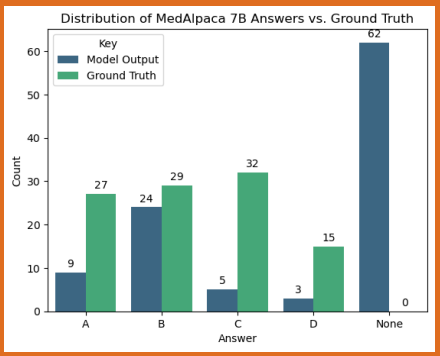
The gains weren't massive, but they were consistent.

Answer Distributions









The train/test sets were balanced, so the evaluation was fair.

Baseline TinyLlaMA was super biased toward one answer choice.

Fine-tuning helped spread the answers more evenly.

MedAlpaca did better than the baseline but often skipped questions entirely.

Future Directions

- Scale to larger models
- Improve MCQ generation
- Extend to open-ended QA
- Test on broader datasets

If I had more time, I'd fine-tune larger models like LLaMA-2 and evaluate against GPT models.

I'd also look at smarter ways to generate MCQs, not just rejection sampling.

Lastly, I'd also explore open-ended QA and broader evaluation datasets.

Thank You!

Thank you for listening!